

Worked Example: Kota Padang

Step-by-step calculation of climate impact, cost, fossil energy and land take per tonne of MSW, with uncertainty and sensitivity analysis — MSW pathway model v3.2, baseline year 2025

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Summary for Kota Padang (central values unless stated)

- 2025 tonnage 656.0 t/day; composition sampled in 2023 and used as a 2025 proxy; bulk moisture 0.52; LHV 7.19 MJ/kg; methane potential 45.8 kg CH₄/t (MCF = 1); fossil CO₂ if burned 434 kg/t.
- GHG per tonne: open dump 990, sanitary landfill 561, WtE 160, RDF 345, AD 389, PHB 547, RDF + AD 180 kg CO₂e/t.
- Net cost per tonne: landfill 18.4, WtE 57.9, RDF 38.7, AD 36.6, PHB 38.1, RDF + AD 44.8 USD/t.
- At a carbon value of 50 USD/t the lowest carbon-inclusive cost is SL landfill + flare; RDF + AD overtakes the landfill at 69 USD/t CO₂e.
- Monte Carlo (4,000 draws per fixed carbon value): the landfill is best with probability 0.80 at 50 USD/t; at 100 USD/t RDF + AD with 0.65 and the landfill with 0.21 (standard errors ≤ 0.008). This is decision uncertainty, not sampling noise.
- Sensitivity: landfill-gas collection efficiency and as-received moisture dominate the comparison between landfill and RDF + AD (Sobol total indices 0.31 and 0.25).
- Ex ante: no RDF, AD or WtE plant exists in Padang. Section 10 gives the targets an RDF + AD project would have to meet and the measurements worth making first.

How this document was produced. *docs/padang_calc.py* writes out every equation of the model again with named intermediate quantities and stops if any result differs from the model output (GHG, cost, CED and land take of all options agree to 10⁻⁶). *docs/padang_uncertainty.py* runs the Monte Carlo with the same random numbers as the 21-location analysis, so the probabilities match Table 7 of the methodology. Symbols follow the methodology (Section 4.1).

1 Inputs from the RIPS

Source: RIPS extract (author); RIPS Kota Padang (komposisi domestik/non-domestik terpisah). Composition sampled in 2023 (wet mass, domestic and non-domestic reported separately). Glass is not reported for Padang: the cell is blank and treated as not reported, not as zero. Data-quality flags: C4 glass not reported (blank, not zero); C0 composition is the RIPS sampling-year distribution used as a 2025 proxy, not a 2025 measurement

RIPS category	Domestic %	Non-domestic %	Model fraction
sisa_makanan	43.95	51.56	food
daun_kering	15.68	25.78	garden
kertas_kardus	8.85	6.34	paper
kayu	0.42	0.26	wood
plastik	19.84	9.61	plastic
logam	1.30	1.36	metal
kain	1.80	0.93	textile
karet_kulit	0.27	0.26	rubber
sterofoam	0.22	0.54	plastic
b3	0.18	0.33	other
elektronik	1.09	0.17	other
lain_lain	6.41	2.85	other
kaca	blank	blank	glass
Sum	100.01	99.99	

Table 1: RIPS composition of Kota Padang and its mapping to the ten model fractions.

Tonnage to 2025. The tonnage comes from the RIPS table of the stated year, so it is projected once with the Padang growth rate:

$$Q_{2025} = Q_t(1 + g)^{2025 - t} = 647.57 \times (1 + 0.0065)^2 = 656.02 \text{ t/day} \quad (1)$$

Domestic 620.73 → 628.83 t/day and non-domestic 26.84 → 27.19 t/day (same factor 1.013042). The tonnage does not enter the GHG per tonne; it enters plant cost (economies of scale) and the scale gates.

2 Composition per functional unit

Each stream is normalised to 1 (domestic sum 100.01%, factor 0.99990; non-domestic 99.99%, factor 1.00010) and the two are weighted by their 2025 masses (95.86% / 4.14%):

$$S_j = \frac{M_d S_j^{(d)} + M_n S_j^{(n)}}{M_d + M_n} \Rightarrow \sum_j S_j = 1 \text{ t per t MSW} \quad (2)$$

Fraction	Domestic (mapped %)	Non-domestic (mapped %)	s domestic	s non-domestic	s used (t/t)
food	43.95	51.56	0.4395	0.5157	0.4426
garden	15.68	25.78	0.1568	0.2578	0.1610
paper	8.85	6.34	0.0885	0.0634	0.0875
wood	0.42	0.26	0.0042	0.0026	0.0041
textile	1.80	0.93	0.0180	0.0093	0.0176
rubber	0.27	0.26	0.0027	0.0026	0.0027
plastic	20.06	10.15	0.2006	0.1015	0.1965
metal	1.30	1.36	0.0130	0.0136	0.0130
glass	0.00	0.00	0.0000	0.0000	0.0000
other	7.68	3.35	0.0768	0.0335	0.0750

Table 2: Composition per tonne of MSW. Rules L and W are not triggered for Padang (organics reported separately; dry leaves present).

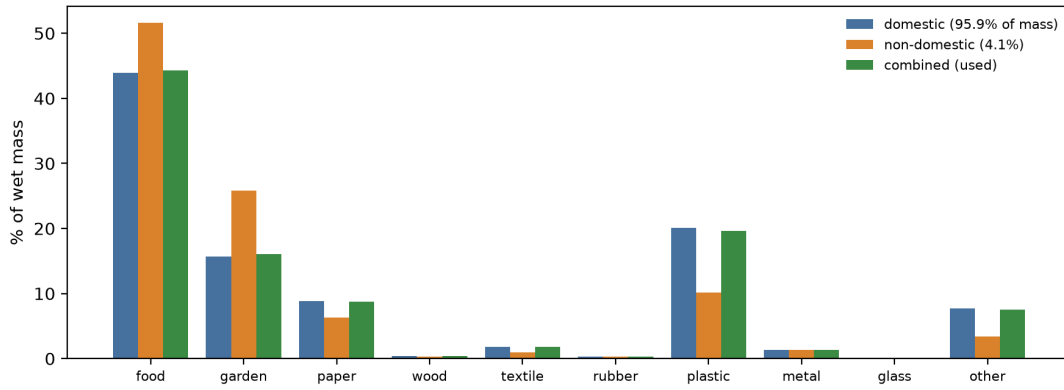


Figure 1: Domestic, non-domestic and combined composition of Kota Padang.

3 Characterisation: dry matter, methane potential, fossil carbon, heating value

Every property multiplies the same dry mass $s_j(1 - w_j)$ (moisture as received, main case). Parameters: Table 2 of the methodology; $\text{DOC} \times \text{DOCf}$ multiplier $k_D = 1.00$, methane fraction $F = 0.50$, heating-value factor $k_h = 0.90$ (legacy fractions; plastic uses its own value), rubber $\text{DOCf} = 0$ (gap scenario).

$$\text{CH}_4 = 10^3 \sum_j S_j (1 - w_j) \text{DOC}_j \text{DOCf}_j k_D F \frac{16}{12}; \quad E_{\text{fos}} = 10^3 \sum_j S_j (1 - w_j) C_j \phi_j \frac{44}{12}; \quad H = \sum_j S_j (1 - w_j) h_j - \lambda \sum_j S_j w_j \quad (3)$$

Fraction	s	w	dry s(1-w)	DOC	DOCf	C	ϕ	h MJ/kg	CH4 kg/t	CO2 fossil kg/t	LHV MJ/kg
food	0.4426	0.75	0.1107	0.38	0.70	0.38	0.00	14.85	19.62	0.00	0.832
garden	0.1610	0.55	0.0724	0.49	0.70	0.49	0.00	14.40	16.56	0.00	0.827
paper	0.0875	0.35	0.0568	0.44	0.50	0.46	0.01	14.40	8.34	0.96	0.744
wood	0.0041	0.25	0.0031	0.50	0.10	0.50	0.00	16.20	0.10	0.00	0.048
textile	0.0176	0.35	0.0115	0.30	0.50	0.50	0.20	16.65	1.15	4.20	0.176
rubber	0.0027	0.15	0.0023	0.47	0.00	0.67	0.20	20.70	0.00	1.13	0.046
plastic	0.1965	0.22	0.1532	0.00	0.00	0.75	1.00	30.50	0.00	421.44	4.569
metal	0.0130	0.05	0.0124	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.002
glass	0.0000	0.02	0.0000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.000
other	0.0750	0.25	0.0562	0.00	0.00	0.03	1.00	0.00	0.00	6.19	-0.046
Total	1.0000	M = 0.521	0.4787						45.77	433.91	7.194

Table 3: Characterisation per tonne of MSW. The LHV column is each fraction's contribution $s(1-w)h - \lambda sw$ (water makes it negative for fractions without heating value).

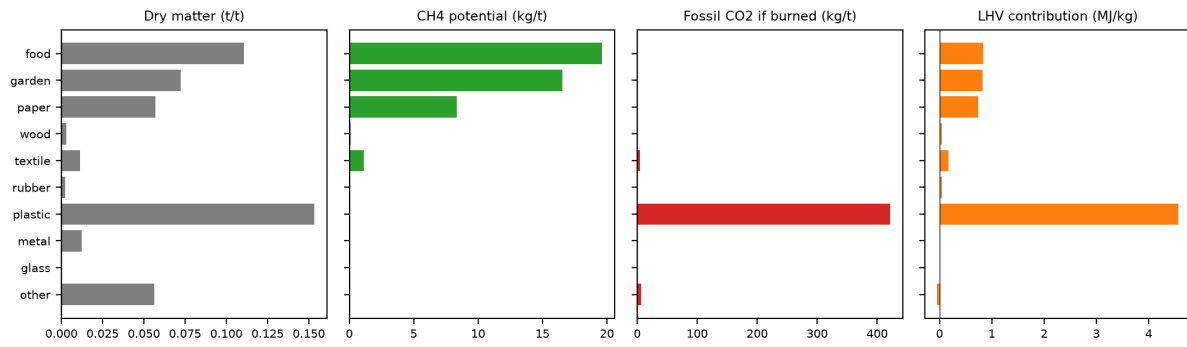


Figure 2: Contribution of each fraction to dry matter, methane potential, fossil CO₂ and heating value.

Food is 44% of the wet mass but, being wet ($w = 0.75$), only 0.111 t/t of dry matter; garden waste (16% of the mass, $w = 0.55$, DOC 0.49) contributes almost as much methane (16.6 vs 19.6 kg/t). Plastic supplies 97% of the fossil CO₂ and 4.57 of the 7.19 MJ/kg. The dry-matter HHV implied is 19.0 MJ/kg, inside the 15.7–19.2 MJ/kg measured for Indonesian MSW (Prabowo et al. 2019).

4 Baselines: open dump and sanitary landfill

Open dump (MCF = 0.8, no gas collection, no oxidation): $CH_4 = 45.77 \times 0.8 = 36.62$ kg/t; $G_{OD} = 36.62 \times 27 + 1$ (diesel) = **989.7 kg CO₂e/t**; cost 4.0 USD/t.

Sanitary landfill (MCF = 1, lifetime gas collection $\eta = 0.50$, flare, cover oxidation 0.10): CH_4 generated 45.77 kg/t, captured and flared 22.89, emitted $(45.77 - 22.89) \times (1 - 0.10) = 20.60$ kg/t; $G_{SL} = 20.60 \times 27 + 5 =$ **561.1 kg CO₂e/t**. Cost: $c_{SL} (Q/500)^{-e} = 20 \times (656.0/500)^{-0.3} =$ **18.44 USD/t**.

The landfill already avoids 429 kg CO₂e/t relative to the open dump, for 14.4 USD/t more.

5 The five recovery options, step by step

Common values: grid Sumatera EF = 0.83 kg CO₂/kWh; nearest kiln Semen Padang (Indarung) at 16.8 km by road; $CRF = r(1+r)^n / ((1+r)^n - 1) = 0.1175$ ($r = 0.10$, $n = 20$); CAPEX per tonne = $K_{ref} (q/0.85/q_{ref})^b CRF / (365q)$ with $b = 0.70$.

5.1 S1 Waste-to-energy

Electricity $E = H \times 1000 / 3.6 \times \eta_w = 7.194 \times 277.8 \times 0.18 = 359.7$ kWh/t.

GHG component	kg CO ₂ e/t	Cost component	USD/t
fossil	433.91	capex	49.10
N ₂ O	13.65	om	27.00
auxiliary	10.00	revenue	-25.18
electricity credit	-299.26	ash landfill	6.99
ash landfill	1.25		
Total G	159.55	Total C	57.91

Table 4: WtE. Fossil CO₂ from Table 3; N₂O 0.05 kg/t $\times$ 273; ash 0.25 t/t to landfill (inert); electricity credited at the grid factor and sold at 0.07 USD/kWh (market case). Gate G1 (LHV ≥ 7 MJ/kg): passed; PSEL tariff ($\geq 1,000$ t/day): not eligible.

5.2 S2 RDF to cement kiln

Fraction	s (t/t)	τ (x k)	to RDF r (t/t)	rejected (t/t)
food	0.4426	0.10	0.0443	0.3984
garden	0.1610	0.40	0.0644	0.0966
paper	0.0875	0.70	0.0612	0.0262
wood	0.0041	0.60	0.0025	0.0017
textile	0.0176	0.70	0.0123	0.0053
rubber	0.0027	0.60	0.0016	0.0011
plastic	0.1965	0.80	0.1572	0.0393
metal	0.0130	0.00	0.0000	0.0130
glass	0.0000	0.00	0.0000	0.0000
other	0.0750	0.10	0.0075	0.0675

Table 5: Sorting of each fraction into RDF (legacy transfer coefficients).

RDF line: mass in 0.3510 t/t with 0.1317 t water and 0.2193 t dry; dried to $\omega = 0.20$: $m_{\text{out}} = \min(0.3510, 0.2193/(1-0.20)) = 0.2741$ t/t. Energy in dried RDF $E = \Sigma r(1-w)h - \lambda(m_{\text{out}} - \text{dry}) = 4.952$ GJ/t; dryer heat (3.2 GJ/t water $\times$ 0.0768 t) leaves $E_{\text{net}} = 4.706$ GJ/t and $m_{\text{del}} = 0.2605$ t RDF/t MSW with NCV = 18.06 MJ/kg (gate G4 $\geq$ 12.56: passed). Coal displaced: $\psi E_{\text{net}} = 0.90 \times 4.706$ GJ at 96.1 kg CO₂/GJ. The rejects (0.6490 t/t, mostly wet food) are landfilled and generate 30.49 kg CH₄/t.

GHG component	kg CO2e/t	Cost component	USD/t
fossil	342.06	capex	11.69
electricity	33.28	om	18.40
diesel	3.00	transport	0.44
transport	0.44	revenue	-5.41
coal credit	-407.04	rejects landfill	13.62
rejects landfill	373.64		
Total G	345.38	Total C	38.74

Table 6: RDF. The methane from rejected wet organics is the largest positive term.

5.3 S3 Anaerobic digestion of food waste

Food captured $a = ks_{\text{food}} = 0.70 \times 0.4426 = 0.3098$ t/t. Methane $V = a \times 1000 \times (1-w_{\text{food}}) \times \text{VS/TS} \times y_{\text{CH}_4} \times k_{\text{mech}} = 0.3098 \times 1000 \times 0.25 \times 0.87 \times 0.36 \times 0.60 = 14.56$ Nm³/t; fugitive 0.05; electricity exported 39.7 kWh/t ($\eta_{\text{CHP}} 0.36$, own use 0.20). The rest of the waste (0.6902 t) and the digestate (0.0620 t) go to the landfill (32.04 kg CH₄/t generated). Front-end separation of food from mixed waste is charged as $\pi = 0.40$ of the RDF line. Food separated mechanically from mixed waste digests less well than source-separated food, hence $k_{\text{mech}} = 0.60$, and needs depackaging and grit removal, charged as $c_{\text{pre}} = 5.0$ USD per tonne of feed (pretreatment in the cost column).

GHG component	kg CO2e/t	Cost component	USD/t
fugitive	14.07	capex	6.32
diesel	0.93	om	4.34
electricity credit	-33.04	pretreatment	1.55
residue landfill	393.02	revenue	-2.78
front end sorting	14.51	residue landfill	15.10
		front end sorting	12.04
Total G	389.49	Total C	36.57

Table 7: AD of food waste separated from mixed waste.

5.4 S4 PHB from landfill gas

Captured methane 22.89 kg/t $\div R = 2.3$ t CH₄/t PHB gives 9.95 kg PHB/t, i.e. 2,383 t PHB/yr at 656 t/day (gate G6 $\geq$ 500 t/yr: passed). Production cost $c_{500} (P/500)^{-e} = 5.97$ USD/kg against a price of 4.0 USD/kg.

GHG component	kg CO2e/t	Cost component	USD/t
landfill CH4	556.15	landfill	18.44
diesel	5.00	PHB production	59.44
PHB chemicals	4.98	PHB revenue	-39.80
PP credit	-18.91		
Total G	547.22	Total C	38.07

Table 8: PHB. Against the landfill it is built on, the only climate gain is the displaced polypropylene.

5.5 S5 Integrated RDF + AD

The whole tonne passes the RDF line, which also separates the food: 0.3098 t to the digester (as in S3), the remaining 0.6902 t sorted as in S2 (RDF 0.2548 t/t, NCV 18.31 MJ/kg); only the rejects (0.3702 t/t) and digestate go to the landfill (18.12 kg CH₄/t).

GHG component	kg CO2e/t	Cost component	USD/t
AD fugitive	14.07	AD capex	6.32
AD diesel	0.93	AD om	4.34
AD electricity credit	-33.04	AD pretreatment	1.55
RDF fossil	342.06	AD revenue	-2.78
RDF electricity	33.28	RDF capex	11.69
RDF diesel	3.00	RDF om	18.40
RDF transport	0.43	RDF transport	0.43
RDF coal credit	-403.39	RDF revenue	-5.36
rejects landfill	222.35	rejects landfill	10.25
Total G	179.68	Total C	44.83

Table 9: RDF + AD. No separate front-end step is charged because the RDF line separates the food; the digester carries the same k_{mech} and c_{pre} as S3.

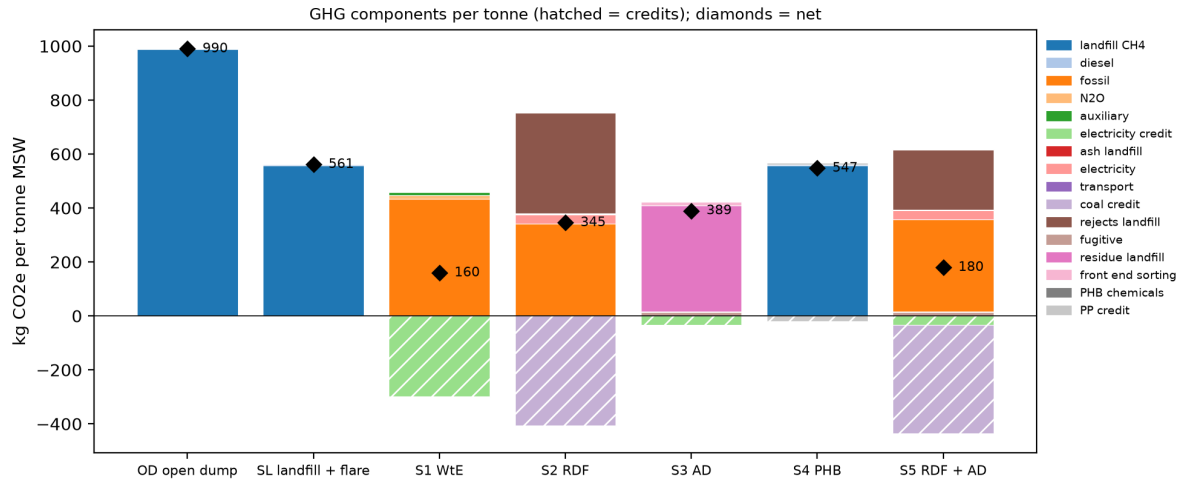


Figure 3: GHG components of each option for Kota Padang.

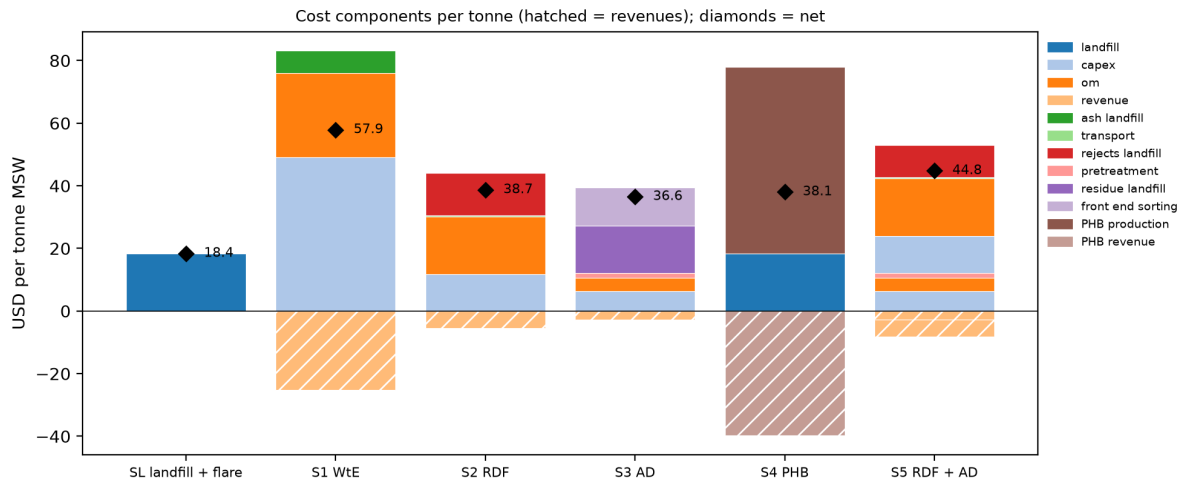


Figure 4: Cost components of each option for Kota Padang.

6 Summary of central results and decision

Option	Gates	G kg CO2e/t	ΔG vs OD	ΔG vs SL	C USD/t	MAC vs SL USD/t	CED MJ/t	Land m2/t
Open dump (baseline)	-	989.7	-	-	4.0	-	-	-
SL landfill + flare	pass	561.1	428.6	-	18.44	-	78	0.0812
S1 WtE	pass	159.5	830.2	401.6	57.91	98	-4,196	0.0189
S2 RDF	pass	345.4	644.3	215.8	38.74	94	-3,858	0.0608
S3 AD	pass	389.5	600.2	171.7	36.57	106	-197	0.0654
S4 PHB	pass	547.2	442.5	13.9	38.07	1410	-580	0.0812
S5 RDF + AD	pass	179.7	810.0	381.5	44.83	69	-4,303	0.0442

Table 10: Central results for Kota Padang (market case, GWP100).

The best option minimises the carbon-inclusive cost $CIC_k = C_k + p G_k/1000$ among the options that pass their gates. Two options with straight lines cross at $p^* = 1000(C_b - C_a)/(G_a - G_b)$: for RDF + AD against the landfill $p^* = 1000 \times (44.83 - 18.44) / (561.1 - 179.7) = 69.2$ USD/t CO₂e.

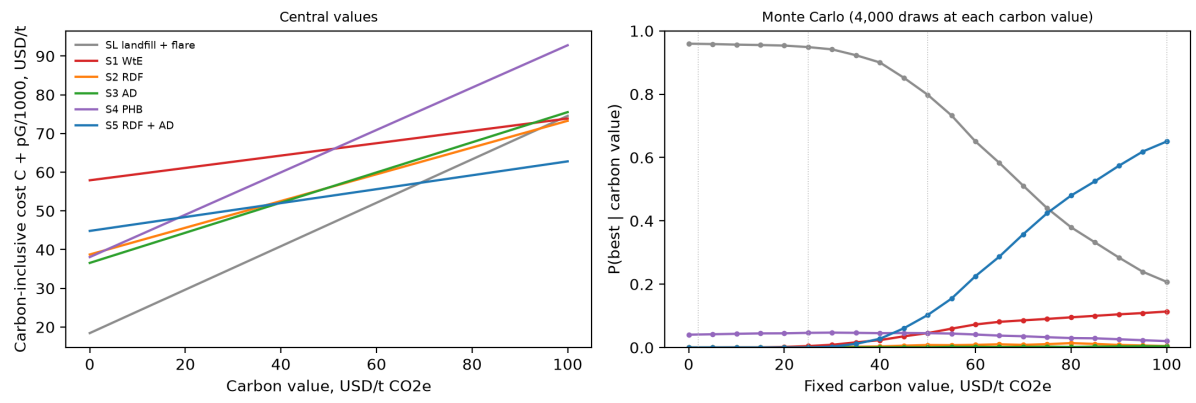


Figure 5: Left: carbon-inclusive cost of each option against the carbon value (central values). Right: probability of being best at each fixed carbon value (Monte Carlo, 4,000 draws; dotted lines at 2, 25, 50 and 100).

7 Uncertainty analysis

4,000 Monte Carlo draws. Composition: Dirichlet around the RIPS proxy with concentration $\alpha_0 = 80$ (heuristic; unflagged data). Parameters: triangular distributions of the register (61 inputs), the realism parameters k_{mech} and c_{pre} and the CED/land-use factors; one common wetness draw moves all moistures. The carbon value is a policy choice and is held fixed.

Fraction	Central	p5	median	p95
food	0.443	0.354	0.441	0.536
garden	0.161	0.099	0.159	0.232
paper	0.087	0.041	0.084	0.141
wood	0.004	0.000	0.001	0.019
textile	0.018	0.002	0.014	0.047
rubber	0.003	0.000	0.000	0.014
plastic	0.196	0.128	0.195	0.275
metal	0.013	0.001	0.009	0.038
glass	0.000	0.000	0.000	0.000
other	0.075	0.033	0.071	0.128

Table 11: Sampled composition (share of wet mass). Bulk moisture p5–p95 0.42–0.60; LHV 5.3–9.8 MJ/kg; WtE heating-value gate passed in 61% of draws.

Option	G p5 / p50 / p95	ΔG vs OD p50 (p5–p95)	C p5 / p50 / p95	CED p50 MJ/t	P(feasible)	P(best) at 50	P(best) at 100
SL landfill + flare	330 / 589 / 926	248 (-72-586)	11.4 / 18.3 / 24.5	87	1.00	0.798 ± 0.012	0.207 ± 0.013
S1 WtE	68 / 180 / 310	656 (329-1059)	44.2 / 63.1 / 85.2	-3,937	0.61	0.045 ± 0.006	0.114 ± 0.010
S2 RDF	188 / 361 / 580	476 (182-815)	31.8 / 39.2 / 46.7	-3,835	1.00	0.008 ± 0.003	0.004 ± 0.002
S3 AD	221 / 408 / 667	430 (147-735)	31.0 / 39.2 / 47.6	-136	1.00	0.001 ± 0.001	0.004 ± 0.002
S4 PHB	313 / 574 / 913	263 (-62-606)	17.6 / 36.3 / 60.9	-451	1.00	0.045 ± 0.006	0.020 ± 0.004
S5 RDF + AD	76 / 191 / 336	645 (364-986)	39.2 / 47.6 / 56.7	-4,262	1.00	0.102 ± 0.009	0.650 ± 0.015

Table 12: Monte Carlo results for Kota Padang (market case). P(best) at a fixed carbon value of 50 or 100 USD/t CO₂e, ± 95% Monte Carlo interval.

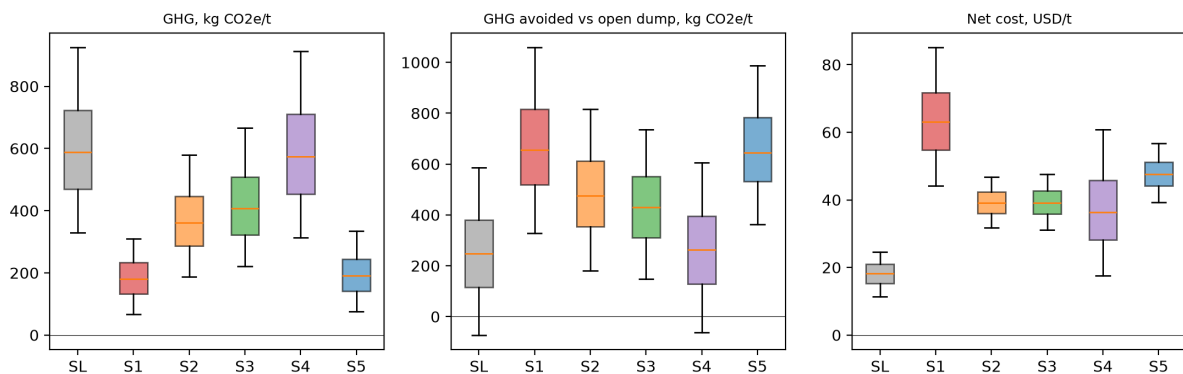


Figure 6: Distributions of GHG, GHG avoided against the open dump and net cost (boxes: interquartile range; whiskers: 5th-95th percentiles).

Indicator	Option	composition only	parameters only	both
C	S1 WtE	10.7	38.9	41.0
C	S2 RDF	4.3	14.1	14.9
C	S3 AD	1.7	16.3	16.6
C	S4 PHB	4.4	43.3	43.2
C	S5 RDF + AD	5.8	16.3	17.5
C	SL landfill + flare	0.0	13.1	13.1
G	S1 WtE	194.6	130.7	242.4
G	S2 RDF	95.9	383.6	392.3
G	S3 AD	181.9	404.5	446.3
G	S4 PHB	161.9	581.2	600.5

Indicator	Option	composition only	parameters only	both
G	S5 RDF + AD	78.6	247.0	259.1
G	SL landfill + flare	166.1	573.4	595.9

Table 13: Width of the 90% interval ($p_{95} - p_5$) when only one source of uncertainty is sampled. G in kg CO₂e/t, C in USD/t.

Composition uncertainty matters most for WtE (through the plastic share: 195 kg CO₂e/t of interval width) and much less for the cost of any option; parameter uncertainty dominates the landfill-based options, mainly through gas collection.

8 Sensitivity analysis

8.1 One-at-a-time (tornado). Each input of the register (and k_{mech} , c_{pre}) is set to its low and its high value with all others at their central values. The decisive output is the carbon-inclusive cost gap between RDF + AD and the landfill at 50 USD/t (base value 7.32 USD/t; negative = RDF + AD cheaper on carbon-inclusive cost).

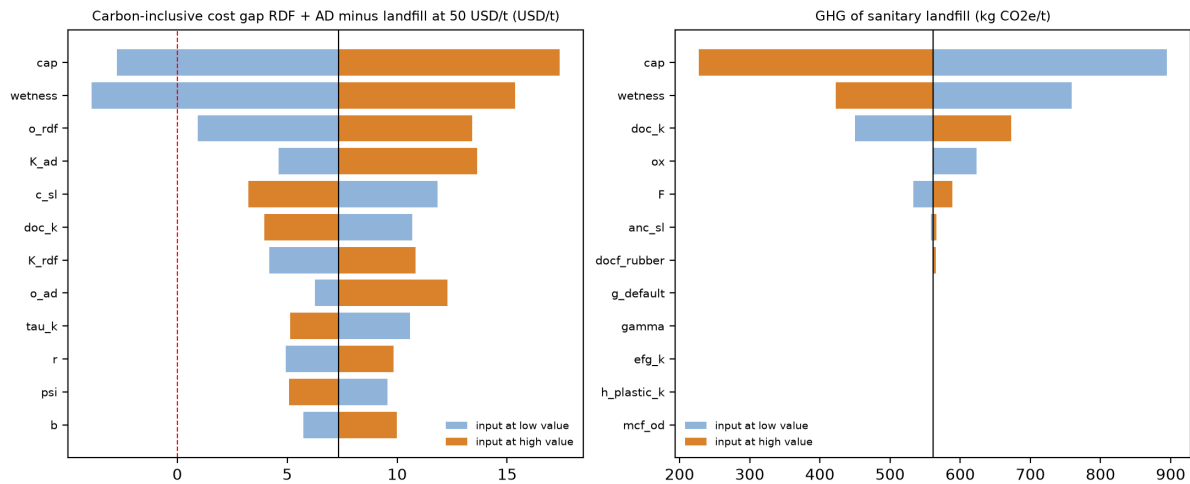


Figure 7: Tornado diagrams for Kota Padang. Left: carbon-inclusive cost gap S5 minus SL at 50 USD/t (red dashed line: break-even). Right: GHG of the sanitary landfill.

Input	Meaning	Range	Gap at low	Gap at high	Crosses zero?
cap	Lifetime landfill-gas collection efficiency, sanitary landfill	0.2 to 0.8	-2.76	17.40	yes
wetness	All fraction moistures at their low / high bound	0 to 1	-3.90	15.36	yes
o_rdf	RDF plant O&M;	12 to 24.5	0.92	13.42	no
K_ad	AD plant CAPEX at reference feed capacity	4 M to 14 M	4.61	13.64	no
c_sl	Full cost of sanitary landfill with gas collection and flare	9 to 30	11.83	3.23	no
doc_k	Multiplier on DOC x DOCf (default uncertainty +-20%)	0.8 to 1.2	10.68	3.96	no
K_rdf	RDF plant CAPEX at reference capacity	9 M to 16 M	4.18	10.84	no
o_ad	AD plant O&M;	10.6 to 30	6.27	12.28	no
tau_k	Multiplier on RDF transfer coefficients	0.85 to 1.1	10.58	5.13	no
r	Real discount rate, same for all pathways	0.08 to 0.12	4.93	9.84	no

Table 14: The ten inputs with the largest swing of the S5 – SL carbon-inclusive cost gap. 'Crosses zero' means that the input alone, within its range, can reverse the choice between the two options.

8.2 Variance-based (Sobol). First-order (S1) and total (ST) indices from 1,024 Saltelli base samples (all register inputs plus the wetness draw varied together; composition fixed at the Padang values).

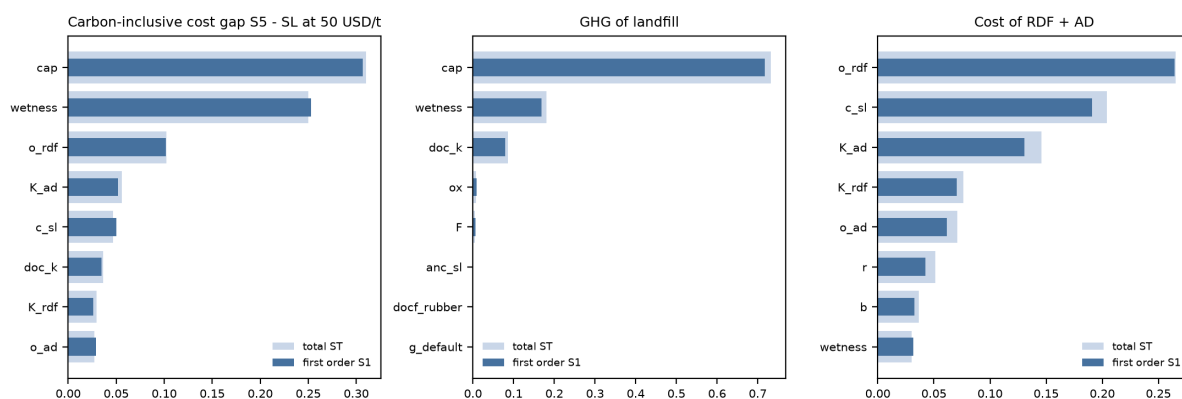


Figure 8: Sobol indices for Kota Padang.

Output	Most influential inputs: ST (S1)
Carbon-inclusive cost gap S5 - SL at 50 USD/t	cap 0.31 (0.31); wetness 0.25 (0.25); o_rdf 0.10 (0.10); K_ad 0.06 (0.05); c_sl 0.05 (0.05)
GHG landfill	cap 0.73 (0.72); wetness 0.18 (0.17); doc_k 0.09 (0.08); ox 0.01 (0.01); F 0.01 (0.01)
GHG WtE	efg_k 0.46 (0.43); eta_wte 0.45 (0.44); h_k 0.05 (0.04); h_plastic_k 0.04 (0.04); wetness 0.02 (0.01)
GHG RDF + AD	cap 0.65 (0.64); kappa 0.08 (0.08); doc_k 0.08 (0.06); psi 0.06 (0.06); tau_k 0.03 (0.03)
Cost WtE	K_wte 0.54 (0.53); p_el 0.10 (0.09); wetness 0.09 (0.09); o_wte 0.08 (0.08); r 0.08 (0.07)
Cost RDF + AD	o_rdf 0.27 (0.26); c_sl 0.20 (0.19); K_ad 0.15 (0.13); K_rdf 0.08 (0.07); o_ad 0.07 (0.06)

Table 15: Sobol indices for Kota Padang.

Option	GHG: top inputs rho	Cost: top inputs rho
SL landfill + flare	cap 0.82; moisture 0.28; doc_k 0.24; s_garden 0.19	c_sl 1.00; cap 0.05; sub_pp 0.04; h_plastic_k 0.04
S1 WtE	s_plastic 0.82; eta_wte 0.36; efg_k 0.32; s_food 0.32	K_wte 0.69; moisture 0.34; p_el 0.31; s_plastic 0.27
S2 RDF	cap 0.84; doc_k 0.25; psi 0.15; moisture 0.13	c_sl 0.64; o_rdf 0.52; K_rdf 0.28; s_plastic 0.24
S3 AD	cap 0.77; s_garden 0.32; moisture 0.29; doc_k 0.22	c_sl 0.62; pre 0.40; K_ad 0.37; o_ad 0.25
S4 PHB	cap 0.83; moisture 0.27; doc_k 0.23; s_garden 0.18	c_phb 0.66; p_phb 0.55; c_sl 0.32; cap 0.25
S5 RDF + AD	cap 0.75; s_garden 0.28; kappa 0.25; psi 0.23	o_rdf 0.44; c_sl 0.41; K_ad 0.34; s_food 0.28

Table 16: Spearman rank correlations in the Monte Carlo sample (composition shares included as s_...).

9 Scenarios

Scenario	Q t/d	Central best at 0	25	50	100	Most probable at 50 (P)	Most probable at 100 (P)
Market	656	SL	SL	SL	S5	SL (0.80)	S5 (0.65)
Perpres 109	656	SL	SL	SL	S5	SL (0.80)	S5 (0.65)
Food sep.	656	SL	S3	S3	S3	S3 (0.82)	S3 (0.51)
AD feed optimistic	656	SL	SL	SL	S5	SL (0.68)	S5 (0.78)
Residues to OD	656	SL	SL	SL	S1	SL (0.61)	S5 (0.50)
Managed M1	n.a.						
Managed M2	471	SL	SL	SL	S5	SL (0.83)	S5 (0.65)
GWP20	656	SL	SL	S1	S1	S5 (0.47)	S1 (0.56)
IPCC moisture	656	SL	SL	S5	S5	S5 (0.37)	S5 (0.67)
Overview projected	656	SL	SL	SL	S5	SL (0.80)	S5 (0.65)
Rubber DOCf 0.5	656	SL	SL	SL	S5	SL (0.80)	S5 (0.65)
Glass imputed	656	SL	SL	SL	S5	SL (0.81)	S5 (0.65)
Pseudocount 0.5	656	SL	SL	SL	S5	SL (0.79)	S5 (0.67)
Moisture high	656	SL	SL	SL	SL	SL (0.96)	SL (0.67)
RDF NCV -22%	656	SL	SL	SL	S5	SL (0.87)	SL (0.43)
PHB large-scale (what-if)	656	S4	S4	S4	S4	S4 (0.99)	S4 (0.80)

Table 17: Scenarios for Kota Padang. Managed M1 is n.a. because Padang has no status-index value; M2 uses the TPA-delivery lower bound (71.8%).

Interpretation for Padang. At the carbon values that apply today the landfill with flare is the clear choice. RDF + AD overtakes it at about 69 USD/t CO₂e at central values, so at 100 USD/t it is the most probable option, but not robustly: with the upper moisture bound the most probable option at 100 USD/t is SL, and with a 22% lower RDF heating value it is SL. A cement kiln 17 km away makes RDF cheap to deliver, and the tonnage (656 t/day) is below the Perpres 109 threshold, so the WtE tariff does not apply. Separating food at source makes S3 the most probable

option at 50 USD/t; optimistic AD feed alone leaves SL in front.

10 What Padang would need: break-even targets and the value of information

Option	Input	Central	Better if	Break-even at 25	at 50	at 100
S5 RDF + AD	y_pen_mech	0.6	higher	never	never	already
S5 RDF + AD	pre_ofmsw	5	lower	never	never	already
S5 RDF + AD	p_rdf	1.15	higher	4.76	2.72	already
S5 RDF + AD	o_rdf	18.4	lower	1.54	11.1	30.2
S5 RDF + AD	cap	0.5	lower	never	0.282	0.675
S3 AD	y_pen_mech	0.6	higher	never	never	0.724
S3 AD	pre_ofmsw	5	lower	never	never	1.88
S3 AD	o_ad	14	lower	never	never	10.9
S3 AD	cap	0.5	lower	never	never	0.471
S2 RDF	p_rdf	1.15	higher	4.32	3.17	0.879
S2 RDF	o_rdf	18.4	lower	3.49	8.89	19.7
S2 RDF	cap	0.5	lower	never	never	0.534
S1 WtE	p_el	0.07	higher	0.152	0.124	0.0681
S1 WtE	K_wte	1.2e+08	lower	4.81e+07	7.26e+07	1.22e+08
S1 WtE	o_wte	27	lower	never	7.6	27.7

Table 18: Break-even values for Kota Padang: the value of one input (others central) at which the option's carbon-inclusive cost equals the landfill's. 'never': no value in the tested range suffices; 'already': the option is cheaper over the whole range. Units as in the register (K_wte in USD).

Measurement group	EVPI at 25	at 50	at 100 (USD/t)	at 100, USD/yr
Landfill gas performance (collection, DOC, F, oxidation)	0.000	0.019	0.562	134,618
Waste characterisation (moisture, composition)	0.000	0.035	0.352	84,223
WtE performance and cost	0.000	0.001	0.261	62,525
RDF line and off-take (O&M, CAPEX, price, coal substitution, sorting)	0.000	0.001	0.001	180
AD on mixed waste (yield, penalty, capture, costs, pre-treatment)	0.000	0.000	0.000	41
Grid and electricity price	0.000	0.000	0.000	0
Landfill cost	0.000	0.000	0.000	0
Finance (discount rate, lifetime, scale exponent)	0.000	0.000	0.000	0
EVPI (all inputs)	0.216	1.151	3.147	753,482

Table 19: Value of information for Kota Padang (USD per tonne and per year at the 2025 tonnage). Group values are conservative and need not add up to the EVPI.

For Padang this means: at 50 USD/t an RDF + AD project would have to secure an RDF processing cost of about 11.1 USD/t or an RDF price of about 2.72 USD/GJ to match the landfill. Perfect information would be worth about 753,482 USD per year at 100 USD/t but only 51,838 USD per year at 25 USD/t, so the case for a local waste characterisation and a landfill-gas measurement grows with the carbon value the city or its funders intend to apply.

Caveats. Composition from 2023 used as a 2025 proxy; moisture as received, dry heating values and RDF transfer coefficients are assumptions; costs are class-5 estimates; the Padang managed share (93.7%) is provisional; land-use and several CED factors are assumptions.